

ULTRON 2.7 Phase 9 Execution Report

Offline-first voice provider stack

Project	ULTRON 2.7
Phase	Phase 9: Offline Voice Stack
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 9 upgrades ULTRON 2.7 from browser-only voice behavior to an offline-first provider stack. The system can now select local STT/TTS adapters, report provider health, fall back gracefully, and keep typed/browser operation available when local models are not installed.

Voice still flows through STT, the brain/runtime, typed tool calls, schema validation, policy gates, the executor, audit logging, response text, and TTS. No voice provider receives raw shell authority or executor authority.

Phase 9 Scope

- Add local STT provider adapters for faster-whisper and whisper.cpp.
- Add local TTS provider adapters for Piper and pyttsx3.
- Keep browser transcript and browser speech synthesis as fallbacks.
- Add voice provider config fields, environment overrides, and provider health checks.
- Expose provider status and provider test API endpoints.
- Update the web UI to display active STT/TTS providers and run provider tests.
- Add mocked provider-selection and health endpoint tests.

What Was Implemented

Area	File(s)	Result
Configuration	src/ultron27/config.py, ultron.config.example.json	Added provider, model path, device, identity, rate, pitch, and volume settings.
Voice runtime	src/ultron27/voice.py	Added provider config, health model, local provider adapters, browser fallback, and mocks.
Local API	src/ultron27/web_server.py	Added provider health and test routes while preserving command and voice routes.

Area	File(s)	Result
Web interface	web/index.html, web/styles.css, web/app.js	Added active provider display, refresh/test controls, and backend-aware browser speech fallback.
Launcher	scripts/run_phase9_voice_ui.py	Adds a Phase 9-named launcher for the local interface.
Docs	README.md, docs/architecture.md, docs/phase9_offline_voice_stack.md	Documents config, provider matrix, health checks, and fallback behavior.
Tests	tests/test_pipeline.py	Expanded to 47 tests covering provider selection, fallback, mocks, and API endpoints.

Provider Matrix

Provider	Type	Behavior
browser	STT	Browser recognition sends transcript text to the backend.
text_payload	STT	Accepts transcript/text JSON payloads.
faster_whisper	STT	Checks for package and model path before activation.
whisper_cpp	STT	Checks for executable and model path before activation.
browser_speech_synthesis	TTS	Delegates audio playback to the browser.
piper	TTS	Checks for Piper and local voice model before activation.
pyttsx3	TTS	Uses the local pyttsx3 package when available.
mock	STT/TTS	Deterministic tests and demos.

Safety Boundary

Boundary	Phase 9 Behavior	Safety Property
STT	Transcribes or accepts text payloads.	Output is treated as a user goal, not executable authority.
Brain/runtime	Routes every goal through the existing system.	Validation, policy, executor, and audit remain authoritative.
High-risk commands	Pause for explicit confirmation.	Destructive actions remain dry-run or not implemented.
TTS	Speaks or delegates response text.	Speech output cannot trigger tools.
Fallbacks	Missing providers downgrade to browser/typed behavior.	Unavailable local models do not crash the app.

Verification Results

Check	Command	Result
Unit tests	<code>\$env:PYTHONPATH='src'; python -m unittest discover -s tests</code>	47 tests passed.
Compile check	<code>python -m py_compile src/ultron27/config.py src/ultron27/voice.py src/ultron27/web_server.py scripts/run_phase9_voice_ui.py</code>	Phase 9 Python modules compiled successfully.
Frontend syntax	<code>bundled node --check web/app.js</code>	Frontend JavaScript parsed successfully.
Dataset evaluation	<code>python scripts/evaluate_dataset.py --split test</code>	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	<code>python scripts/evaluate_safety_regression.py</code>	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
Provider API	ThreadingHTTPServer endpoint tests	Provider status, test-STT, and test-TTS routes returned expected payloads.

Recommended Next Step

- Install a real faster-whisper or whisper.cpp model and verify microphone audio capture end to end.
- Install a Piper voice model or pyttsx3 on the target Windows machine and tune ULTRON's voice identity.
- Add stronger wake-word and VAD layers after the offline providers are stable.